

Exact role-divisor retrodiction in Kerr–Newman thermodynamics via an inverse-channel complementarity metric

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Abstract

For the two-role Kerr thermodynamic surface, the directive–substrate–product (DBP) role–channel calculus yields a diagonal channel–inverse metric whose scalar curvature obeys a complementarity law: it remains finite on the Davies transition locus and diverges on the physical role boundaries, with pole order 3 at a generic boundary and pole order 4 at a reflection–fixed boundary. The corresponding three-role construction is not automatic, since each output chart carries several off-diagonal couplings and the diagonal assembly rule is no longer unique. We resolve this selection problem for the Kerr–Newman surface.

The selected metric is an inverse-channel metric $g_F(u=0)$: for each output role the mass–role coupling channels are inverted before summing, while the role–role channels are omitted. Within the natural weighted family, this is the unique choice that is both positive definite and free of interior curvature poles. The omitted role–role channels instead detect interior channel–isotropy strata.

On the outer physical wedge, $g_F(u=0)$ is rational in (S, J, Q, π) . Its scalar curvature has no interior singular support, including on the Davies surface; its singular support in the closure lies exactly on the role divisors $T = 0$, $\Omega = 0$, $\Phi_e = 0$. The Kerr complementarity law lifts to Kerr–Newman: the extremal divisor is generic and has order 3, while the two reflection–fixed role faces (spin $\Omega = 0$ and electric charge $\Phi_e = 0$) have order 4. The leading reflection coefficients are obtained in closed form, and the extremal coefficient is shown to be strictly negative at every open point of the extremal edge—so the extremal pole has exact order 3 throughout—by proving strict transverse monotonicity, $\partial_S(G_J G_Q)|_{S=S_{\text{ext}}} < 0$, via a full-edge coefficient–positivity and Sturm certificate. All algebraic identities are checked by reproducible symbolic certificates.

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1 Introduction

The $n = 2$ picture. Thermodynamic geometry equips a fundamental relation with a Riemannian metric whose curvature diagnoses phase structure—the Weinhold [1] and Ruppeiner [2] information metrics, the Legendre-invariant program of geometrothermodynamics (GTD) [4], and, as a complementary reading, the heat-capacity (Davies [3]) transition loci. The directive-substrate-product (DBP) role-channel calculus [5] is one such reading. Given a fundamental relation $M = \Phi(E^1, \dots, E^n)$ read as a hypersurface in role variables, it assigns to each output chart $E^i = f_i(M, E^{j \neq i})$ a family of coupling channels, and, at $n = 2$, a diagonal metric

$$h = \sum_i \kappa_{c,i} dE^i \otimes dE^i, \quad g^{\text{DBP}} = -h^{-1}, \quad (1)$$

where $\kappa_{c,i} = -\Lambda_i^2/q_i^2$ is the (single) coupling channel of chart i . For the Kerr surface this metric’s scalar curvature satisfies a complementarity law: it is finite on the Davies curve and blows up on the two physical role boundaries (extremal $T = 0$ and zero-spin $\Omega = 0$), with pole order 3 on the generic boundary and 4 on the reflection-fixed one. We use this $n = 2$ result as the reference complementarity law, re-deriving below only the ingredients needed for the Kerr–Newman lift.

The $n = 3$ selection problem. At $n = 3$ the recipe (1) is under-determined: each output chart carries *several* off-diagonal couplings, and there is no unique way to assemble them into a single diagonal entry. This is the calculus’s stated open problem: *does a three-role DBP complementarity metric exist that lifts the Kerr law?*

Result. We answer in the affirmative for the Kerr–Newman surface and select the metric uniquely. Write the couplings in each output chart as two *mass–role* couplings $\Lambda_{i,\{M,j\}}$ (joining the graph value M to a role variable) and one *role–role* coupling $\Lambda_{i,\{j,k\}}$. The metric is

$$g_F(u=0)_{ii} = q_i^2 \sum_{\text{mass-role } \rho} \frac{1}{\Lambda_{i,\rho}^2}, \quad (2)$$

i.e. one inverts each mass–role channel *before* summing, and *omits* the role–role channel. Here “role” is used in the role–coordinate sense: the extensive state variables (S, J, Q) , not only the electromagnetic charge Q . Our contributions:

1. **Selection (Thm. 4).** Among the natural assembly rules, and within the weighted invert-then-sum family $g_F(u)$, the value $u = 0$ is the unique choice that is positive definite *and* free of interior curvature poles. The omitted role–role couplings carry the interior channel-isotropy strata.

2. **Rationality and regularity (Thms. 5, 8).** On the outer physical wedge W_+ the metric (2) is rational in (S, J, Q, π) , and its only singular support lies on the role divisors $U_S J Q = 0$, i.e. $T = 0$, $\Omega = 0$, $\Phi_e = 0$.
3. **Complementarity law and exact coefficients (Thm. 18).** The scalar curvature has an order-3 pole on the generic extremal edge and order-4 poles on the two reflection-fixed role faces; the reflection coefficients are $-14/B$ for an explicit collapsing coefficient B , given in closed form, and the sign of the extremal coefficient is fixed by a transverse monotonicity.
4. **A general normal form and the corner (Props. 10, 17).** The $-14/B$ is the $m = 2$ case of a parity-fixed inverse-channel normal form with scalar curvature $-m(m+5)x^{-4}/B$ ($m =$ transverse fibre dimension). Where the two order-4 faces meet—the Schwarzschild corner—the balanced approach cancels them back to order 2, the mildest point of the boundary, along a Newton wedge $m(a) = \max(4a - 2, 4 - 2a)$.

Reader map. The argument has three steps. First, the metric is *selected* by excluding interior channel-isotropy poles (§3). Second, *rationality* reduces the singular support to the explicit mass-role coupling factors (§4). Third, two local Laurent normal forms determine the pole orders and leading coefficients (§§5–6).

Relation to GTD. The construction below is not proposed as a Legendre-invariant Quevedo metric, nor as a replacement for Weinhold- or Ruppeiner-type pullback geometries. It is a role-chart diagnostic: its test is retrodictive, namely whether the scalar curvature recovers the known thermodynamic role divisors and distinguishes their local symmetry type. The result should be read as a complementarity metric within the DBP role calculus, not as a claim of uniqueness among all thermodynamic geometries.

Verification. Every exact algebraic assertion below is checked by a reproducible symbolic certificate (`verification/lead7_testk*.py`); numerical curvature is used only as an independent consistency check after the exact derivation. The diagonal 3D Ricci harness is validated on flat $\mathbb{R}^3$ ($R = 0$) and the round three-sphere ($R = 6/a^2$).

2 Set-up

Fundamental relation and role coordinates. The Kerr–Newman fundamental relation, with the area normalisation $S = A/4$, is $M^2 = U(S, J, Q)$ with

$$U(S, J, Q) = \frac{S}{4\pi} + \frac{\pi J^2}{S} + \frac{Q^2}{2} + \frac{\pi Q^4}{4S}. \quad (3)$$

The role coordinates are the extensive variables (S, J, Q) (entropy, angular momentum, electric charge); $M = \sqrt{U}$ is the graph value. The first derivatives are

$$U_S = \frac{S^2 - \pi^2(4J^2 + Q^4)}{4\pi S^2}, \quad U_J = \frac{2\pi J}{S}, \quad U_Q = \frac{Q(S + \pi Q^2)}{S}. \quad (4)$$

The conjugate potentials are $T = \partial M / \partial S = U_S / 2M$, $\Omega = U_J / 2M$, $\Phi_e = U_Q / 2M$.

Definition 1 (Outer physical wedge).

$$W_+ = \{S, J, Q > 0, U_S > 0\} = \{S, J, Q > 0, S^2 > \pi^2(4J^2 + Q^4)\}.$$

Equivalently W_+ is the outer-horizon branch

$$S = S_+ := \pi(2M^2 - Q^2 + 2\sqrt{\text{disc}}), \quad \text{disc} := M^4 - M^2Q^2 - J^2 > 0,$$

on which $T > 0$; the inner branch is $S = S_- := \pi(2M^2 - Q^2 - 2\sqrt{\text{disc}})$.

Remark 2 (Why W_+ and not $\{\text{disc} > 0\}$). The region $\{\text{disc} > 0\}$ also contains the inner branch $S = S_-$, on which $U_S < 0$; there some of the positivity below fails. For instance at $(S, J, Q) = (\pi\sqrt{3}/2, \frac{1}{4}, 1)$ one has $\text{disc} = \frac{1}{48} > 0$ but $U_S = -\frac{1}{6\pi} < 0$, and a coupling numerator vanishes. All theorems are stated on W_+ .

By (4), on W_+ the three role divisors are cut out by a single product:

$$T = 0 \iff U_S = 0, \quad \Omega = 0 \iff J = 0, \quad \Phi_e = 0 \iff Q = 0. \quad (5)$$

Definition 3 (Davies surface). The Davies surface referred to below is the fixed- (J, Q) heat-capacity divergence locus,

$$\partial_S T|_{J,Q} = 0 \iff 2U U_{SS} - U_S^2 = 0,$$

which lies in the interior of W_+ for the relevant parameter range. It serves here as an interior comparison locus—the transition where the $n = 2$ information/GTD curvatures diverge—not as a role divisor.

3 The metric and the selection theorem

Couplings. For an output chart with output role i and the other two roles $\{a, b\}$, define the chart norm and the role-coupling numerators by

$$q_i = 1 + |\nabla f_i|^2 = 1 + \frac{4U + U_a^2 + U_b^2}{U_i^2} = \frac{U_i^2 + 4U + U_a^2 + U_b^2}{U_i^2}, \quad D_{i,a} = U_{ii}U_a - U_iU_{ia}, \quad (6)$$

$$\Lambda_{i,\{M,a\}} = \frac{2M D_{i,a}}{U_i^3}, \quad \text{so} \quad \Lambda_{i,\{M,a\}}^2 = \frac{4U D_{i,a}^2}{U_i^6}. \quad (7)$$

The mass-role pairs $\{M, a\}$ use $D_{i,a}$. The role-role pair $\{a, b\}$ uses the mixed second partial of the output chart, obtained from $M^2 = U$ by implicit differentiation at fixed M :

$$\Lambda_{i,\{a,b\}} = \frac{\partial^2 f_i}{\partial E_a \partial E_b} = -\frac{D_{i,\{a,b\}}}{U_i^3}, \quad D_{i,\{a,b\}} = U_i^2 U_{ab} - U_i(U_a U_{ib} + U_b U_{ia}) + U_a U_b U_{ii}. \quad (8)$$

This channel enters $g_F(u)$ only through the weight u and is discarded at $u = 0$ (Appendix A); its only appearance below is that its interior zero locus is a channel-isotropy stratum (Prop. 7). The three output charts of (3) are solved explicitly in Appendix A.

The family $g_F(u)$. The two natural ways to assemble a diagonal entry from the two mass–role channels and the one role–role channel are

$$\underbrace{g_{ii} = \left(\sum_{\rho} C_{i,\rho} \right)^{-1}}_{\text{sum-then-invert}} \quad \text{vs} \quad \underbrace{g_{ii} = \sum_{\rho} C_{i,\rho}^{-1}}_{\text{invert-then-sum}}, \quad C_{i,\rho} = \frac{\Lambda_{i,\rho}^2}{q_i^2}.$$

The invert–then–sum form with a role–role weight u is

$$g_F(u)_{ii} = q_i^2 \left(\frac{1}{\Lambda_{i,\{M,j\}}^2} + \frac{1}{\Lambda_{i,\{M,k\}}^2} + \frac{u}{\Lambda_{i,\{j,k\}}^2} \right). \quad (9)$$

Both assembly rules reduce to (1) at $n = 2$ (a single mass–role pair, no role–role pair), so all satisfy the $n = 2$ reduction constraint.

Theorem 4 (Selection of $u = 0$). *Sum–then–invert fails the complementarity law: its scalar curvature is finite on the reflection–fixed faces $\Omega = 0$ and $\Phi_e = 0$ for every role–role weight. Invert–then–sum with $u > 0$ reproduces the boundary divergences with the correct orders 3/4/4 but develops a spurious interior curvature pole on the locus $\Lambda_{Q,\{S,J\}} = 0$, which lies in the interior of W_+ and is distinct from the Davies surface. The choice $u = 0$ removes the interior pole while retaining all boundary divergences, and is the unique member of (9) that is both positive definite and interior–regular on W_+ : any $u > 0$ develops a pole at the role–role zero, any $u < 0$ loses positive definiteness there.*

Certificate–assisted proof. The reflection–face divergences come from mass–role couplings that vanish on the face ($\Lambda_{S,\{M,J\}} \propto J$, etc., Prop. 6); the role–role coupling of the Q –chart vanishes on an interior curve (Prop. 7). Sum–then–invert averages the collapsing channel against the finite ones, so the metric entry stays finite where the reflection channel alone would force a pole—the curvature never sees it (verified symbolically and by high–precision curvature; `lead7_test3`). Invert–then–sum keeps each channel’s zero as a metric pole; the mass–role zeros are the role boundaries (desired), but the role–role zero is interior (undesired). Near that zero $\Lambda_{Q,\{S,J\}} \rightarrow 0$, so $g_F(u)_{QQ} \sim u/\Lambda^2$ for $u > 0$; the curvature certificate (`lead7_test3`) verifies that this metric pole produces a scalar–curvature pole. For $u < 0$ the same term drives the diagonal entry negative near the zero, so positive definiteness is lost; only $u = 0$ is finite and positive there. The boundary divergences are u –independent because they live in the mass–role sector (`lead7_test6/7`). $\square$

Henceforth $g_F := g_F(0)$, with diagonal entries

$$G_i := g_F ii = q_i^2 \left(\Lambda_{i,\{M,j\}}^{-2} + \Lambda_{i,\{M,k\}}^{-2} \right) = \frac{(U_i^2 + 4U + U_a^2 + U_b^2)^2 U_i^2}{4U} \left(\frac{1}{D_{i,a}^2} + \frac{1}{D_{i,b}^2} \right), \quad (10)$$

the second equality being the radical–free form obtained from (6)–(7) (the $M = \sqrt{U}$ cancels). The numerator carries q_i^2 , hence the term $(U_i^2 + 4U + U_a^2 + U_b^2)^2$ with the leading U_i^2 : this is the graph normalization $q_i = 1 + |\nabla f_i|^2$ of (6), *not* $|\nabla f_i|^2$. The $+U_i^2$ is essential—it is what makes G_i reduce to the $n = 2$ channel metric $g_{ii} = q_i^2/\Lambda_i^2$ and it fixes the transverse values G_J, G_Q that enter the extremal coefficient (Prop. 16); dropping it changes C_{ext} . (On a collapsing channel $U_i \rightarrow 0$ makes $q_i \rightarrow \infty$, so $q_i \sim |\nabla f_i|^2$ and the $+U_i^2$ is invisible in the reflection coefficient A_2 and C_Ω, C_Φ ; it survives only in the finite transverse channels.)

4 Rationality and interior regularity

Theorem 5 (Rationality). *On W_+ the metric entries G_S, G_J, G_Q of (10), and hence the scalar curvature $R[g_F]$, are rational functions of (S, J, Q, π) .*

Proof. The couplings (7) contain M only through $\Lambda^2 \propto M^2 = U$; the norm q_i^2 likewise carries $4M^2 = 4U$. Substituting $M^2 = U$ into (10) removes all radicals, giving the displayed rational form in the polynomial data U, U_i, U_{ij} . The diagonal Ricci scalar of a diagonal metric is a rational function of the entries and their partials. Certificate `lead7_test7` (T7-b) checks the rational form agrees with the radical construction. $\square$

The six mass–role numerators factor explicitly:

$$\begin{aligned} D_{S,J} &= \frac{J(4\pi^2 J^2 + \pi^2 Q^4 + S^2)}{2S^4}, & D_{S,Q} &= \frac{Q(4\pi^2 J^2 Q^2 + 8\pi J^2 S + \pi^2 Q^6 + 2\pi Q^4 S + Q^2 S^2)}{4S^4}, \\ D_{J,S} &= \frac{4\pi^2 J^2 - \pi^2 Q^4 + S^2}{2S^3}, & D_{J,Q} &= \frac{2\pi Q(S + \pi Q^2)}{S^2}, \\ D_{Q,J} &= \frac{2\pi J(S + 3\pi Q^2)}{S^2}, & D_{Q,S} &= \frac{(S + \pi Q^2)^3 - 4\pi^2 J^2(S + 3\pi Q^2)}{4\pi S^3}. \end{aligned} \quad (11)$$

Proposition 6 (Mass–role zeros occur only at role divisors). *On W_+ every mass–role coupling is finite and nonzero. In the closure $\overline{W_+}$ the zero locus of each mass–role coupling, and the polar locus of each, is contained in $\{J = 0\} \cup \{Q = 0\} \cup \{U_S = 0\}$.*

Proof. On W_+ one has $U_S > 0$: indeed the horizon Vieta identity gives $U_S = \sqrt{\text{disc}}/S_+$. Using $U_S = [S^2 - \pi^2(4J^2 + Q^4)]/(4\pi S^2)$, each numerator in (11), after clearing the positive power of S , is a polynomial with nonnegative coefficients in S, J, Q, π and the nonnegative quantity U_S . Hence: $D_{S,J}, D_{Q,J} \propto J$ vanish only on $\Omega = 0$; $D_{S,Q}, D_{J,Q} \propto Q$ only on $\Phi_e = 0$. For the remaining two, $U_S > 0$ gives, respectively, $4\pi^2 J^2 - \pi^2 Q^4 + S^2 > 8\pi^2 J^2 > 0$ and $(S + \pi Q^2)^3 > 4\pi^2 J^2(S + 3\pi Q^2)$, so $D_{J,S}, D_{Q,S}$ vanish only where $U_S = 0$ jointly with $J = 0$ or $Q = 0$. Generic $U_S = 0$ enters as the native *pole* (denominator U_i^3), not a numerator zero. So no mass–role coupling vanishes in the open interior. Certificate `lead7_test4`. $\square$

Proposition 7 (Role–role interior zero). *By (8) with $U_{JQ} = 0$, the Q –chart role–role coupling is $\Lambda_{Q,\{S,J\}} = -D_{Q,\{S,J\}}/U_Q^3$ with*

$$D_{Q,\{S,J\}} = -\frac{J(12\pi^3 J^2 Q^2 + 4\pi^2 J^2 S + 3\pi^3 Q^6 + 5\pi^2 Q^4 S + \pi Q^2 S^2 - S^3)}{2S^4}.$$

The bracket has indefinite sign (through the $-S^3$ term), so $\Lambda_{Q,\{S,J\}}$ vanishes on an interior curve of W_+ . On the $M = 2$ slice this occurs at $J \approx 0.826$, $Q \approx 1.699$ (with $S = S_+ \approx 28.266$, so $\text{disc}/M^4 \approx 0.236$); on the same M, J slice the Davies point occurs at $Q \approx 1.656$. The two loci are distinct: the role–role zero is a channel–isotropy stratum, not a role boundary.

Theorem 8 (Interior regularity). *$R[g_F]$ is finite and analytic on the interior of W_+ ; its only singular support in $\overline{W_+}$ is $U_S J Q = 0$, the three role divisors.*

Proof. By Prop. 6 each $\Lambda_{i,\{M,a\}}$ is finite and nonzero on the open wedge, so each G_i in (10) is a positive analytic function there, hence $\det g_F = \prod_i G_i \neq 0$. Therefore g_F^{-1} , the Christoffel symbols and the curvature are finite analytic functions on the interior. Singular support can occur only where some G_i degenerates or blows up, i.e. where a mass–role coupling hits 0 or ∞ ; by Prop. 6 that is contained in $\{J = 0\} \cup \{Q = 0\} \cup \{U_S = 0\}$. The role–role zeros of Prop. 7 do not enter because $u = 0$ omits those channels. Certificate `lead7_test4`. $\square$

5 Two local Laurent lemmas

The pole orders and coefficients follow from two exact local models, verified symbolically (certificates `lead7_test6/7`).

Lemma 9 (Quadratic coordinate collapse with finite transverse components). *Let a diagonal metric near $x = 0$ have the native component collapsing quadratically and the two transverse components finite,*

$$ds^2 = (A_2x^2 + A_3x^3 + O(x^4))dx^2 + (P_0 + P_1x + O(x^2))dy^2 + (R_0 + R_1x + O(x^2))dz^2,$$

with $A_2, P_0, R_0 \neq 0$. Then

$$R = \frac{1}{A_2} \left(\frac{P_1}{P_0} + \frac{R_1}{R_0} \right) x^{-3} + O(x^{-2}),$$

independent of A_3 and all higher data. The odd order arises from the linear drift P_1, R_1 of the transverse components.

The reflection order and its universal coefficient are a special case of a general normal form for a parity-fixed inverse-channel geometry: one collapsing coordinate against a flat transverse fibre.

Proposition 10 (Parity-fixed inverse-channel normal form). *Let $m \geq 1$, $B > 0$, $A_\alpha > 0$. For the pure diagonal normal form on an $(m+1)$ -dimensional state space,*

$$ds^2 = Bx^2 dx^2 + x^{-2} \sum_{\alpha=1}^m A_\alpha dy_\alpha^2,$$

the scalar curvature is exactly

$$R = -\frac{m(m+5)}{B} x^{-4},$$

independent of all transverse amplitudes A_α . In particular $m = 2$ (a three-dimensional state space) gives $R = -14x^{-4}/B$.

Proof. Set $r = \frac{1}{2}\sqrt{B}x^2$, so $dr = \sqrt{B}x dx$ and $x^{-2} = \sqrt{B}/(2r)$. The metric becomes the warped product $ds^2 = dr^2 + w(r)^2 h_{\text{flat}}$ with flat fibre $h_{\text{flat}} = \sum_{\alpha} A_\alpha dy_\alpha^2$ and $w(r)^2 = \sqrt{B}/(2r)$, i.e. $w \propto r^{-1/2}$. For a warped product over a flat m -dimensional fibre, $R = -2m w''/w - m(m-1)(w'/w)^2$; with $w'/w = -1/(2r)$ and $w''/w = 3/(4r^2)$,

$$R = -\frac{6m}{4r^2} - \frac{m(m-1)}{4r^2} = -\frac{m(m+5)}{4r^2},$$

and $4r^2 = Bx^4$ gives the claim. The two contributions are radial focusing ($6m$) and transverse fibre shear ($m(m-1)$). (Verified independently from a first-principles Ricci computation for $m = 1, 2, 3$: $R = -6, -14, -24$ times x^{-4}/B ; certificate `lead7_test10`.) $\square$

Corollary 11 (Asymptotic parity-fixed face). *Suppose near $x = 0$, with all components even in x ,*

$$g_{xx} = B(y)x^2 + O(x^4), \quad g_{\alpha\alpha} = A_\alpha(y)x^{-2} + O(1), \quad B(y) > 0, \quad A_\alpha(y) > 0.$$

Then $R = -\frac{m(m+5)}{B(y)} x^{-4} + O(x^{-2})$; for $m = 2$, $R = -14x^{-4}/B(y) + O(x^{-2})$.

Remark 12 (The transverse x^{-2} poles are essential). *The collapsing-only form $ds^2 = Bx^2 dx^2 + dy^2 + dz^2$ is flat ($R \equiv 0$, after $r = \sqrt{B}x^2/2$): the $m = 0$ member has no fibre. The order-4 pole needs a genuinely collapsing fibre, i.e. the transverse x^{-2} poles. The DBP reflection faces carry them (§6), so Cor. 11 applies with $m = 2$; it promotes the generic order 3 to the reflection order 4, and makes the coefficient $-14/B$ inevitable rather than a symbolic coincidence.*

6 The complementarity law and exact coefficients

6.1 Reflection faces $\Omega = 0$ and $\Phi_e = 0$

On approach to $J = 0$ the metric is even in J with the native J -component collapsing and the transverse S, Q components diverging as J^{-2} (the parity-fixed normal form, Cor. 11 with $x = J$, $m = 2$). Writing $G_J = B_J(S, Q)J^2 + O(J^4)$,

$$B_J = \frac{D_{\Omega,1}D_{\Omega,2}^2}{256Q^2S^5\pi^3(S-\pi Q^2)^2}, \quad D_{\Omega,1} = (S - \pi Q^2)^2 + 16\pi^2Q^2S^2, \quad D_{\Omega,2} = D_{\Omega,1} + 16\pi S^3, \quad (12)$$

so that $C_\Omega = -14/B_J$, i.e.

$$C_\Omega(S, Q) = -\frac{3584Q^2S^5\pi^3(S-\pi Q^2)^2}{D_{\Omega,1}D_{\Omega,2}^2}. \quad (13)$$

An analogous computation with $x = Q$ gives

$$C_\Phi(S, J) = -\frac{14336J^2S^5\pi^5(S-2\pi J)^2(S+2\pi J)^2(4\pi^2J^2+S^2)}{D_{\Phi,1}D_{\Phi,2}^2}, \quad (14)$$

with $D_{\Phi,1} = (S^2 - 4\pi^2J^2)^2 + 64\pi^4J^2S^2$ and $D_{\Phi,2} = D_{\Phi,1} + 64\pi^3J^2S^3 + 16\pi S^5$.

Proposition 13. $D_{\Omega,1}, D_{\Omega,2} > 0$ and $D_{\Phi,1}, D_{\Phi,2} > 0$ for $S, J, Q > 0$. Hence $C_\Omega < 0$ on $\Omega = 0$ away from the corner $S = \pi Q^2$, and $C_\Phi < 0$ on $\Phi_e = 0$ away from the corner $S = 2\pi J$; at those corners—the intersections $\Omega = 0 \cap T = 0$ and $\Phi_e = 0 \cap T = 0$ —the coefficient vanishes to second order.

6.2 The extremal edge $T = 0$

On approach to $S_{\text{ext}} = \pi\sqrt{4J^2 + Q^4}$ along $\delta = S - S_{\text{ext}}$, only the native S -component collapses ($G_S = A_2\delta^2 + O(\delta^3)$) while G_J, G_Q stay finite, so Lemma 9 applies (with $x = \delta$). The collapsing coefficient is

$$A_2 = \frac{(4U + U_J^2 + U_Q^2)^2}{4U} \left(\frac{1}{U_J^2} + \frac{1}{U_Q^2} \right) \Big|_{S=S_{\text{ext}}} > 0, \quad (15)$$

the U_S^2 term of q_S dropping since $U_S = 0$ at extremality. That this closed form is *identically* the δ^2 -Taylor coefficient of G_S is certified symbolically (`lead7_test7`); the U_{SS} cancels between $(U_S/\delta)^2$ and $D_{S,\cdot} = U_{SS}U$, which is why A_2 is free of A_3 and of U_{SS} . Positivity is immediate: (15) is a sum of positive terms. The extremal coefficient is then

$$C_{\text{ext}} = \frac{1}{A_2} \left(\frac{P_1}{P_0} + \frac{R_1}{R_0} \right) = \frac{1}{A_2} \partial_S \log(G_J G_Q) \Big|_{S=S_{\text{ext}}}. \quad (16)$$

Because G_J, G_Q and A_2 are rational (Thm. 5) and the extremal surface is $J^2 = (S^2 - \pi^2Q^4)/(4\pi^2)$, C_{ext} is a rational function of (S, Q, π) ; its denominator factors, up to a positive constant, as

$$\begin{aligned} \text{den} &\propto Q^2\pi(3Q^2\pi + S)(Q^4\pi + Q^2S - Q^2\pi + S)(Q^4\pi^2 + 4Q^2S^2\pi^2 - 2Q^2S\pi + S^2) \\ &\quad \times (Q^4\pi^2 + Q^2S\pi - Q^2\pi^2 + 2S^2 + S\pi)^3 \\ &\quad \times (Q^{10}\pi^3 + Q^8S\pi^2 - 9Q^6S^2\pi^3 + 3Q^4S^3\pi^2 + 5Q^2S^4\pi + S^5). \end{aligned} \quad (17)$$

Proposition 14 (Extremal denominator positivity). *Every factor of (17) is positive on the open edge $S > \pi Q^2$. In particular, with $t = S/(\pi Q^2) > 1$ the last factor equals $Q^{10}\pi^3[1 + t + \pi^2 t^2 h(t)]$ with $h(t) = t^3 + 5t^2 + 3t - 9$; since $h(1) = 0$ and $h'(t) = 3t^2 + 10t + 3 > 0$, one has $h(t) > 0$ for $t > 1$, so the bracket is a sum of positive terms.*

Remark 15 (Interlock). *The degree-ten factor of (17) is exactly the numerator factor carried by $R_0 = G_Q|_{S_{\text{ext}}}$ (certificate `lead7_test7`): the same object whose positivity Prop. 14 settles via h controls both the extremal denominator and the transverse metric value $P_0, R_0 > 0$ that Lemma 9 needs. The order-3 conclusion is thus entirely symbolic: Lemma 9 (the δ^{-3} normal form) together with $A_2 > 0$ (15) and $C_{\text{ext}} < 0$ (Prop. 16) give exact order 3 at every open point, with no fitted exponent.*

6.3 Extremal transverse monotonicity

The coefficient C_{ext} has a geometric sign form. Since $A_2 > 0$ (eq. (15)) and $P_0 = G_J|_{S_{\text{ext}}} > 0$, $R_0 = G_Q|_{S_{\text{ext}}} > 0$ (metric entries), (16) gives $\text{sgn } C_{\text{ext}} = \text{sgn}(P_1/P_0 + R_1/R_0) = \text{sgn } W$, where—differentiating in the ambient entropy $\tilde{S}$ at fixed (J, Q) and then eliminating J on the extremal edge—

$$W(S, Q) := \left[\partial_{\tilde{S}}(G_J G_Q)(\tilde{S}, J, Q) \right]_{\tilde{S}=S, J^2=(S^2-\pi^2 Q^4)/(4\pi^2)}, \quad S > \pi Q^2. \quad (18)$$

Thus whether the order-3 extremal pole can cancel is exactly the question of whether the transverse metric product $G_J G_Q$ (the role-frame transverse density) has a critical point under outward entropy displacement from extremality.

Proposition 16 (Transverse monotonicity). *On the open extremal edge, written as $Q > 0, S > \pi Q^2$ after eliminating J , one has $W(S, Q) < 0$. Consequently $C_{\text{ext}} < 0$ there, and the extremal pole has exact order 3 at every open point.*

Certificate-assisted proof. By (18) it suffices to show $L := \partial_{\tilde{S}} \log(G_J G_Q)|_{S_{\text{ext}}} < 0$ on the open edge. Parametrise the edge by $q = Q^2 > 0$ and $t = S/(\pi Q^2) > 1$; there $L = -P/D$ for polynomials P, D in (q, t, π) . Both $P > 0$ and $D > 0$ are certified by coefficient positivity: after extracting positive monomial factors in π, q, t and applying the substitutions $t = 1 + r$ ($r > 0$) and $\pi^2 = 9 + b$ ($b > 0$, the rational bound $\pi > 3$), each remaining factor is a *nonzero* polynomial in the positive variables q, r, b with nonnegative coefficients, hence strictly positive on the open edge. The denominator D factors into nine such coefficient-positive factors. The numerator is quadratic in q , $P = Aq^2 + Bq + C$; the coefficients B, C are coefficient-positive, and after removing a positive monomial $A = C_0 \pi^2 t^2 (\pi^2 F + G)$ with $C_0 > 0$ and

$$F(t) = (t-1)t^3(t+3)^3 \tilde{H}(t), \quad \tilde{H}(t) = 2t^5 + 5t^4 - 15t^3 - 7t^2 + 15t + 8.$$

A Sturm count gives $\tilde{H} > 0$ on $[1, \infty)$, so $F > 0$ for $t > 1$; a second Sturm count gives $9F + G > 0$ on $[1, \infty)$. Since $\pi^2 > 9$ and $F > 0$,

$$\pi^2 F + G = (\pi^2 - 9)F + (9F + G) > 0,$$

hence $A > 0$. Therefore $P > 0$, and with $D > 0$ we get $L = -P/D < 0$; thus $C_{\text{ext}} < 0$ on the whole open edge. Certificate `lead7_test8_extremal_gate_graphnorm` verifies all factor and Sturm subcertificates reproducibly. $\square$

6.4 The double-reflection corner

The two charge faces $\Omega = 0$ and $\Phi_e = 0$ meet on the Schwarzschild ray $J = Q = 0$ ($S = s$ free). Approach it along the wedge $Q = \epsilon$, $J = \rho \epsilon^a$ ($\rho > 0$, $a \geq 0$, $\epsilon \rightarrow 0^+$). Write $N_0(s) = 4U_0 + U_{S,0}^2 = s/\pi + 1/(16\pi^2)$ for the corner value of the chart norm ($U_0 = s/4\pi$, $U_{S,0} = 1/4\pi$). On the balanced diagonal $a = 1$ the metric entries have the closed-form limits

$$\epsilon^6 G_S \rightarrow \frac{N_0^2 s^5}{\pi(s + 8\pi\rho^2)^2} = \frac{(16\pi s + 1)^2 s^5}{256\pi^5(s + 8\pi\rho^2)^2}, \quad G_J \rightarrow \frac{N_0^2 \pi \rho^2}{s}, \quad G_Q \rightarrow \frac{N_0^2 s}{4\pi\rho^2}, \quad (19)$$

so G_S collapses as ϵ^6 while G_J, G_Q are ϵ -independent and scale as $\rho^{\pm 2}$ (their product $G_J G_Q \rightarrow N_0^4/4$ is ρ -independent). The $1/(16\pi^2) = U_{S,0}^2$ inside N_0 is the graph-norm term; dropping it gives the schematic $s^7/[\pi^3(s + 8\pi\rho^2)^2]$, low by the factor $(1 + 1/(16\pi s))^2$ (≈ 1.04 at $s = 1$).

Proposition 17 (Corner order and Newton wedge). *Along $Q = \epsilon$, $J = \rho \epsilon^a$ the scalar curvature diverges as $R \sim \epsilon^{-m(a)}$ with*

$$m(a) = \max(4a - 2, 4 - 2a),$$

a Newton polygon with vertex $(a, m) = (1, 2)$ and edges $m = 4 - 2a$ ($0 \leq a \leq 1$, the Φ_e side, $m(0) = 4$) and $m = 4a - 2$ ($a \geq 1$, the Ω side). The minimum over all directions is $m = 2$, attained only on the balanced diagonal: $\epsilon^2 R \rightarrow \mathcal{R}_0(s, \rho) < 0$, finite. The double-reflection corner is thus the mildest point of the boundary—the two order-4 collapses partially cancel to order 2.

The corner value $\mathcal{R}_0(s, \rho)$ is controlled at its own extremes by the reflection coefficients: their corner residues are

$$\kappa_\Omega(s) = \lim_{Q \rightarrow 0} \frac{C_\Omega}{Q^2} = -\frac{3584\pi^3 s}{(16\pi s + 1)^2}, \quad \kappa_\Phi(s) = \lim_{J \rightarrow 0} \frac{C_\Phi}{J^2} = -\frac{14336\pi^5}{s(16\pi s + 1)^2},$$

with $\rho^4 \mathcal{R}_0 \rightarrow \kappa_\Omega(s)$ as $\rho \rightarrow 0$ and $\mathcal{R}_0/\rho^2 \rightarrow \kappa_\Phi(s)$ as $\rho \rightarrow \infty$. But $\mathcal{R}_0$ is *not* the two-face superposition: $\mathcal{R}_0(s, \rho) - \kappa_\Omega(s)\rho^{-4} - \kappa_\Phi(s)\rho^2$ is a nonzero mixed-channel term (numerically $\mathcal{R}_0(1, 1) = -1801.25$ against the superposition -1711.56 , an excess -89.70), the curvature signature of the cross term in G_S 's denominator that neither single face resolves. All of (19), Prop. 17, the residues, and the mixed excess are in certificate `lead7_test9` (exact-partial curvature; the κ 's are exact residues of the closed forms C_Ω, C_Φ).

Theorem 18 ($n = 3$ complementarity law). *On the open wedge W_+ , $R[g_F]$ is finite and analytic, including on the Davies surface. As one approaches the boundary role divisors in $\overline{W}_+$, the scalar curvature has the leading behaviours*

$$R \sim C_{\text{ext}} \delta^{-3} \ (T = 0), \quad R \sim C_\Omega J^{-4} \ (\Omega = 0), \quad R \sim C_\Phi Q^{-4} \ (\Phi_e = 0),$$

C_{ext} rational in (S, Q, π) with positive denominator and (Prop. 16) $C_{\text{ext}} < 0$ on the open edge, and C_Ω, C_Φ given by (13)–(14).

7 Discussion

What selects the metric. The result says the DBP complementarity metric is inverse-channel geometry built only from the couplings that carry the role-boundary valuation, *excluding* the role-role couplings, whose zeros are interior channel-isotropy strata belonging to a different (isotropy) diagnostic. Sum-then-invert is too coarse: it hides the reflection zeros. Full invert-then-sum is too sharp: it develops poles on the isotropy strata. The selected object is therefore neither an information/pullback metric nor the naive channel-norm inverse.

Why 3/4/4. The order law is structural. The extremal face is a quadratic native collapse with finite transverse channels (Lemma 9), giving order 3. Each reflection–fixed role face is a parity–fixed face whose transverse channels carry x^{-2} poles—one collapsing coordinate against a flat 2–dimensional fibre. By the parity–fixed normal form (Prop. 10, Cor. 11) the scalar curvature is $-m(m+5)x^{-4}/B$ with $m = 2$ here, i.e. $-14/(Bx^4)$, promoting the order from 3 to 4. The coefficient $14 = 6 \cdot 2 + 2 \cdot 1$ splits into radial focusing and transverse fibre shear; the amplitudes drop out because they only rescale a flat fibre. The two order–4 faces then meet at the Schwarzschild corner, where the balanced approach cancels back to order 2 (Prop. 17)—the mildest, not the wildest, point of the boundary.

A The output charts and the role–role channel

Solving $M^2 = U(S, J, Q)$ for each role gives the output branches (the outer horizon branch for S , the positive role branch for J and Q):

$$\begin{aligned} S &= \pi \left(2M^2 - Q^2 + 2\sqrt{M^4 - M^2Q^2 - J^2} \right), \\ J &= \sqrt{M^4 - M^2Q^2 - w^2}, \quad w := \frac{S}{2\pi} - M^2 + \frac{Q^2}{2}, \\ Q &= \sqrt{-\frac{S}{\pi} + \frac{2}{\pi}\sqrt{\pi SM^2 - \pi^2 J^2}}. \end{aligned}$$

These formulae are used only to identify the output charts; the channel coefficients themselves are more cleanly computed by implicit differentiation of $M^2 = U$, as in (7) and (8).

For output role i with remaining roles $\{a, b\}$, the role–role channel is the mixed second partial $\partial^2 f_i / \partial E_a \partial E_b$; by implicit differentiation it is rational in the second jet of U (eq. (8), verified against the explicit chart mixed partials in certificate `lead7_test4`). In the weighted family $g_F(u)$ this channel appears only through the role–role weight u , and at the selected value $u = 0$ it is discarded. Its interior zero locus is therefore not a thermodynamic role divisor; it is the channel–isotropy stratum of Prop. 7, cut out for the Q –chart by the indefinite factor displayed there.

B Certificates

All exact claims are machine–checked and reproducible: `lead7_test3` (selection / candidate comparison), `lead7_test4` (mass–role zeros = role divisors; Thm. 8), `lead7_test5` (orders, high precision), `lead7_test6` (reflection coefficients C_Ω, C_Φ), `lead7_test7` (generic δ^{-3} lemma, A_2 identically the δ^2 –coefficient of G_S , rational C_{ext} , denominator, interlock), `lead7_test8_extremal_gate_graphnorm` (full–edge extremal transverse monotonicity $W < 0$ for the graph–normalized metric: $L_{\text{ext}} = -P/D$ with D a product of nine coefficient–positive factors, $P = Aq^2 + Bq + C$ with B, C coefficient–positive and $A > 0$ by two Sturm certificates ($\tilde{H}$ of degree 5 and $9F + G$); Prop. 16), `lead7_test9` (double–reflection corner: closed–form metric limits, Newton wedge $m(a) = \max(4a - 2, 4 - 2a)$, corner residues, mixed excess; Prop. 17), `lead7_test10` (parity–fixed normal form $R = -m(m+5)/Bx^{-4}$ from first–principles Ricci for $m = 1, 2, 3$; Prop. 10). All were run twice with identical output.

References

- [1] F. Weinhold, *Metric geometry of equilibrium thermodynamics*, J. Chem. Phys. **63** (1975) 2479.

- [2] G. Ruppeiner, *Riemannian geometry in thermodynamic fluctuation theory*, Rev. Mod. Phys. **67** (1995) 605; erratum **68** (1996) 313.
- [3] P. C. W. Davies, *The thermodynamic theory of black holes*, Proc. R. Soc. Lond. A **353** (1977) 499.
- [4] H. Quevedo, *Geometrothermodynamics*, J. Math. Phys. **48** (2007) 013506.
- [5] W. Lloyd, *Two geometric readings of a thermodynamic surface: the directive–substrate–product role–channel calculus*, note (2026).